Numerical processing of financial data
APM 50442 EP
Part 1: Sequential stock data in one dimension

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1. Sequential stock data in one dimension
   - Around prices and returns
   - Volatility
   - Serial dependence
   - Some stylized facts on returns
2. Sequential stock data in higher dimension
3. From sequential data to portfolio selection
4. Market microstructure

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Some definitions

- **Financial asset**: It is a non-physical asset that represents a claim to future cash flows or ownership rights. It thus produces a certain gain for its holder and it involves a certain risk. It is generally transferable and negotiable on a financial market. It typically includes instruments like stocks, bonds, derivatives...
- **Financial market**: It is a marketplace where buyers and sellers trade financial assets, such as stocks, bonds, commodity-related contracts, currencies... It can operate through exchanges (formal and generally regulated place/system, such as NYSE, Euronext) or over-the-counter (OTC) networks.
- **(Stock) index**: it is a statistical measure that tracks the performance of a group of stocks representing a particular segment of the market, such as a specific industry, country, or economic sector. It is used to gauge the overall health and trends of the market or a subset of it. Example: S&P 500, which tracks 500 large U.S. companies.
- **Portfolio**: A pool of assets, with quantities that may vary over time if the trader transacts or if certain assets mature.

Price

What is a price?

- Forward interpretation (fundamental value): Expected value of the present value of all future cash flows (dividends for a stock).
- Market price/spot price: Value of the last transaction.
- In a time interval, typically one day: open, high, low, close prices.
- For daily time series, generally: close price.

Adjusted close useful for single-name stocks

It makes it possible to correct retrospectively the time series of close prices, mainly in case of:

- payment of a dividend (example: close of 100€ on Monday, announcement of a dividend of 4€ after the market closes: adjusted close price is $100 - 4 = 96$€),
- stock split (announcement of a split $2:1$, meaning each investor receives an extra share for each share already owned: adjusted close price of $100/2 = 50$€).

$\rightsquigarrow$ Python exercise, Question 1.

Price returns

Prices are not stationary: difficult to use them in a statistical perspective. We prefer price returns. $\rightsquigarrow$ Python exercise, Question 2.

Two definitions

For a spot price P_t and times $s < t$:

- **arithmetic price return**: $r_{s,t} = \frac{P_t - P_s}{P_s}$,
- **logarithmic price return**: $r_{s,t}^{\log} = \log\left(\frac{P_t}{P_s}\right)$.

- The arithmetic return is the most relevant one economically.
- Justification of the log-return:
  1. In the arithmetic approach, we have $P_{t+\tau} = P_t(1+R)$, with R a random return.
  2. If R modelled by a Gaussian variable $\mathcal{N}(0, \sigma^2\tau)$, the price may become negative (the larger τ, the stronger the probability).
  3. To avoid this, write: $P_{t+\tau} = P_t \exp(R')$.

  $\rightsquigarrow$ Show that we need $\mathbb{E}[R'] \neq 0$ to have $\mathbb{E}[P_{t+\tau}|P_t] = P_t$.
- For small price variations, a Taylor expansion shows that $r_{s,t}^{\log}$ is close to $r_{s,t}$.
  $\rightsquigarrow$ Python exercise, Question 3.

Price returns of a portfolio/index

We consider a portfolio with N assets. For security i, at time t, we observe:

- its market price: $A_{i,t}$;
- its price return compared to time $s < t$: $r_{i,s,t} = \frac{A_{i,t} - A_{i,s}}{A_{i,s}}$;
- its **quantity** (number of shares) in the portfolio: $q_{i,t}$ (be careful, we do not buy ten bonds at the same price as ten options);
- its **weight** in the portfolio (the weight varies over time even without transactions):

$$w_{i,t} = \frac{\text{Market value of security } i \text{ in the portfolio}}{\text{Market value of the portfolio}} = \frac{q_{i,t} A_{i,t}}{\sum_{j=1}^{N} q_{j,t} A_{j,t}}.$$

Price and return of the portfolio

The value of the portfolio is: $P_t = \sum_{j=1}^{N} q_{j,t} A_{j,t}$.
$\rightsquigarrow$ Show that, when there is no transaction, the price return of the portfolio is the weighted average of the price returns of its constituents.

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Instantaneous volatility

- The **(instantaneous) volatility** measures the magnitude of price returns: standard deviation of the distribution of the price return of time t.
- Not observed, so we simply calculate an average volatility over a (small) time interval.
- Two main methods:
  1. Historical volatility: Quadratic variation (moment method, equivalent to maximum-likelihood estimator when returns are Gaussian).
  2. Parkinson volatility: generalized moment method.

Historical and realized volatility (1/2)

Historical volatility

For price returns between s and $t > s$, $r_{s,t}$, which are iid (some possibility to relax the independence assumption), with $\mathbb{E}[r_{s,t}^2] = \sigma_{t-s}^2 < \infty$, we define the estimator:

$$\widehat{\sigma_{n,\tau}^2} = \frac{1}{n}\sum_{i=1}^{n}\left(r_{(i-1)\tau,i\tau} - \frac{1}{n}\sum_{j=1}^{n} r_{(j-1)\tau,j\tau}\right)^2$$

and $\widehat{\sigma_{n,\tau}} = \sqrt{\widehat{\sigma_{n,\tau}^2}}$.

- $\widehat{\sigma_{n,\tau}}$ is a strongly consistent estimator of σ_τ.
  $\rightsquigarrow$ A similar result for $\widehat{\sigma_{n,\tau}^2}$ is straightforward, but why do we have such a result for $\widehat{\sigma_{n,\tau}}$?
- Asymptotic normality: $\sqrt{n}\left(\widehat{\sigma_{n,\tau}} - \sigma_\tau\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{\sigma_\tau^2}{2}\right)$.
  $\rightsquigarrow$ Same question.

Historical and realized volatility (2/2)

Annualization

The time unit in finance is often considered to be the year (except in high frequency), so people appreciate annualizing their volatility (and so consider σ_1, simply written σ). A simple way to do that:

$$\widehat{\sigma}_n = \frac{1}{\sqrt{\tau}} \widehat{\sigma_{n,\tau}}$$

(take $\tau = 1/252$ for daily returns). It requires

- identically distributed returns,
- independent returns (no relaxation of this assumption),
- Gaussian returns.

Realized volatility is a historical volatility computed with intraday price returns (all contained in the same day): no spurious serial dependence of the volatility series.
$\rightsquigarrow$ Python exercise, Question 4.

Parkinson volatility (1/2)

Is it possible to have a better estimate of the volatility than the realized volatility?
Yes, if we use the data inside each time interval (not extensively, but the traditional high and low).

Noting $O_{s,t}$, $H_{s,t}$, and $L_{s,t}$ the open, high, and low prices in the time interval $[s,t]$, the **log high-low range** is

$$d_{s,t} = h_{s,t} - l_{s,t},$$

where

$$\left\{ \begin{array}{lcl} h_{s,t} & = & \ln H_{s,t} - \ln O_{s,t} \\ l_{s,t} & = & \ln L_{s,t} - \ln O_{s,t}. \end{array} \right.$$

Parkinson (1980) proved that, for a Gaussian dynamic of log-prices (iid increments of duration $\to 0$):

$$\left\{ \begin{array}{lcl} \mathbb{E}[d_{t-1,t}] & = & \sqrt{\frac{8}{\pi}}\sigma \\ \mathbb{E}[d_{t-1,t}^2] & = & 4\ln(2)\sigma^2 \\ \mathbb{E}[d_{t-1,t}^4] & = & 9\zeta(3)\sigma^4, \end{array} \right.$$

where $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$ is Riemann's zeta function, with $\zeta(3) \approx 1.202$.

Parkinson volatility (2/2)

Two **Parkinson volatility** estimators (for example $\tau = 1/252$):

$$\widehat{D_{1,n}} = \frac{1}{n\sqrt{\tau}} \sum_{i=1}^{n} d_{(i-1)\tau, i\tau} \sqrt{\frac{\pi}{8}}$$

and

$$\widehat{D_{2,n}} = \left(\frac{1}{n\sqrt{\tau}} \sum_{i=1}^{n} d^2_{(i-1)\tau, i\tau} \frac{1}{4\ln(2)} \right)^{1/2}.$$

- $\widehat{D_{1,n}}$ and $\widehat{D_{2,n}}$ are strongly consistent estimators of σ.
  $\rightsquigarrow$ A simple theorem makes it possible to conclude this, assuming iid returns.
- Asymptotic normality:

$$\begin{cases} \sqrt{n}\left(\widehat{D_{1,n}} - \sigma\right) & \xrightarrow{d} & \mathcal{N}\left(0, \sigma^2\left(\frac{\pi \ln(2)}{2} - 1\right)\right) \\ \sqrt{n}\left(\widehat{D_{2,n}} - \sigma\right) & \xrightarrow{d} & \mathcal{N}\left(0, \frac{\sigma^2}{4}\left(\frac{9\zeta(3)}{16(\ln 2)^2} - 1\right)\right). \end{cases}$$

  $\rightsquigarrow$ Easy for $\widehat{D_{1,n}}$, but why do we also get such a result for $\widehat{D_{2,n}}$?
- Variance of the estimator $\widehat{D_{1,n}}$ thus 0.18 times the variance of realized volatility (for a same number of time intervals).
  $\rightsquigarrow$ Python exercise, Question 5.

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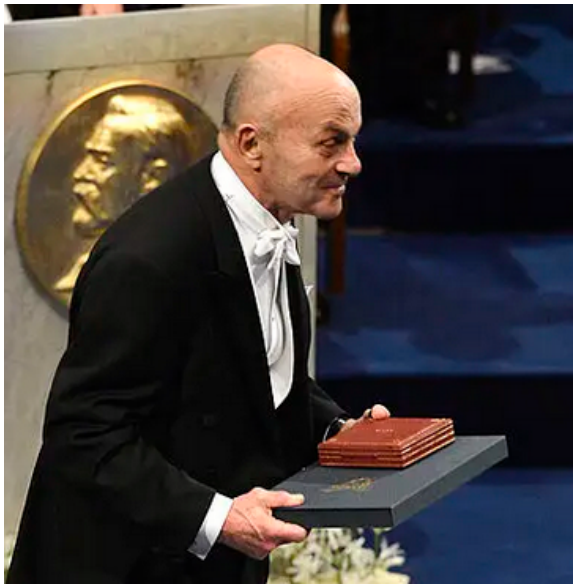
Market efficiency



- Efficient market hypothesis (EMH): *"A market in which prices always fully reflect available* **information** *is called efficient"* (Fama, 1970, Nobel prize 2013).
- Three forms:
  - weak: the information set is just historical prices,
  - semi-strong form: includes public information (announcements of annual earnings, stock splits, etc.),
  - strong form: includes private information for some investors.

  In this presentation, we focus on the weak-form efficiency, particularly on **serial dependence**.

Beyond market efficiency

"The phenomenology of financial markets is incredibly rich and complex, and EMH is only, at best, a medieval flat earth approximation" (Jean-Philippe Bouchaud).

- The logical implications of EMH:
  - Investors are rational (what about behavioural finance?).
  - Prices are martingales (what about statistical arbitrage?).
  - The securities are substitutable for the same level of risk.
- Numerous alternatives to the EMH: fractal market hypothesis, noisy market hypothesis, etc., including the attractive **adaptive market hypothesis** (Andrew Lo, 2004):
  - A statistical model may lead to a performing investment strategy.
  - But the market participants adapt to this strategy (and are subject to natural selection as well). The strategy only performs for a limited time.
  - Conclusion: EMH only works for long time scales.

Measuring market efficiency

- Since the EMH is controversial, one has to quantify and test it.
- Objective: determining market regimes (can one make any forecast?). Useful at least for:
  - quantitative portfolio managers, to help them define a universe of predictable assets to trade,
  - market makers, to help them adjust their prices and avoid informed trades.

Among the several methods for measuring efficiency:

- Autocorrelation (linear model).
- An almost model-free approach: information theory.

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Autocorrelation

The autocorrelation function $\tau \mapsto \gamma(\tau)$ of the series of returns (R_t) is defined as follows:

$$\gamma(\tau) = \mathrm{Corr}(R_{t+\tau}, R_t) = \frac{\mathbb{E}[R_{t+\tau}R_t] - \mathbb{E}[R_{t+\tau}]\mathbb{E}[R_t]}{\left(\mathrm{Var}(R_{t+\tau})\mathrm{Var}(R_t)\right)^{1/2}}$$

for a given time lag τ.
We can estimate $\gamma(\tau)$ from the observed time series, by replacing expectations with empirical means. Of course, we can consider arithmetic returns or logarithmic ones.

Pearson's correlation test

For a correlation γ and its estimate based on n observed pairs $\hat{\gamma}_n$, we define a test with:

(H_0) $\gamma = 0$,

(H_1) $\gamma \neq 0$.

Let the test statistic be

$$T_n = \frac{\hat{\gamma}_n}{\sqrt{(1-\hat{\gamma}_n^2)/(n-2)}}.$$

Under (H_0), if the pairs are Gaussian, then T_n follows a student distribution with $n-2$ degrees of freedom.

Asymptotic version: Student's distribution then converges toward $\mathcal{N}(0,1)$.
$\rightsquigarrow$ Python exercise, Question 6.

Rank autocorrelation (1/2)

Is serial dependence limited to autocorrelation?

Correlation is a measure of linear dependence. Two dependent variables may be uncorrelated.

$\rightsquigarrow$ Find a theoretical example.

Dependence as comonotonicity

Let $f_{\tau,n}$ be the rank function, so that $f_{\tau,n}(\min R_t) = 1$, $f_{\tau,n}(\max R_t) = n$, and

$$f_{\tau,n}^{-1}(1) \leq f_{\tau,n}^{-1}(2) \leq \ldots \leq f_{\tau,n}^{-1}(n).$$

Spearman's correlation is the linear correlation of ranks:

$$\rho(\tau) = \mathrm{Corr}(f_{\tau,n}(R_{t+\tau}), f_{\tau,n}(R_t)) = \frac{\mathbb{E}[f_{\tau,n}(R_{t+\tau})f_{\tau,n}(R_t)] - \mathbb{E}[f_{\tau,n}(R_{t+\tau})]\mathbb{E}[f_{\tau,n}(R_t)]}{\left(\mathrm{Var}(f_{\tau,n}(R_{t+\tau}))\mathrm{Var}(f_{\tau,n}(R_t))\right)^{1/2}}.$$

Rank autocorrelation (2/2)

Spearman's correlation test

For a correlation ρ and its estimate based on n observed pairs $\hat{\rho}_n$, we define a test with:

(H_0) $\rho = 0$,

(H_1) $\rho \neq 0$.

Let the test statistic be

$$T_n = \frac{\hat{\rho}_n}{\sqrt{(1-\hat{\rho}_n^2)/(n-2)}}.$$

Under (H_0), T_n follows a student distribution with $n-2$ degrees of freedom (no Gaussian assumption required).

Again, when $n \to \infty$, we use the more convenient asymptotic Gaussian distribution.
$\rightsquigarrow$ Python exercise, Question 7.

We would like to have a more general dependence measure.

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A short introduction to information theory

A ball is hidden in one among N boxes. We must find it and ask questions to do so. The *price of information* H is the number of questions to be asked (imagine one pays 1€ for each question).

1. All the boxes are equiprobable: if $N = 1$, then $H = 0$ (trivial); if $N = 2$, then ask one question, $H = 1$; if $N = 4$, $H = 2$ (questions by dichotomy); more generally, $H = \log_2(N)$.
2. Non-equiprobable boxes:
   - k different possible colors for the boxes (n_1 boxes of the first color, ..., n_k boxes for the k-th color). Each color has a probability $p_k = n_k/N$.
   - With the information $I_j =$ *"the ball is in a box of the j-th color"*, the number of questions to be asked is $\log_2(n_j)$. Without information I_j, the cost is $\log_2(N)$. So the price of information I_j is $H(I_j) = \log_2(N) - \log_2(n_j) = -\log_2(p_j)$.
   - Since the color is not necessarily the j-th one, the event I_j has only a probaility p_j and the price of the color information is:

$$H = \mathbb{E}\left[H(I_j)\right] = -\sum_{j=1}^{k} p_j \log_2(p_j).$$

This quantity is called **Shannon's entropy**.

Information theory in finance

The knowledge of a probability distribution provides some information about the value of a random variable to be drawn:

- The minimum uncertainty is for a degenerated distribution (all the probability in one point); it corresponds to the minimum entropy, $H = 0$.
- The maximum uncertainty is provided by a uniform distribution among N values; maximum entropy: $H = \log_2(N)$.

How it is used in finance to measure complexity and efficiency (what is the information contained in a historical distribution?):

- permutation entropy (Bandt and Pompe, 2002): quantifies the uncertainty of the permutation between a series of consecutive prices and its sorted version,
- entropy of a series of 0s (negative price return) and 1s (positive price returns) (Risso, 2008).

From prices to Shannon's entropy...

- From a time series of $n+1$ prices $P_0, ..., P_n$, we build a binary sequence $X_1, ..., X_n$:

$$X_i = \mathbb{1}_{\{P_i - P_{i-1} > 0\}}.$$

- We assume that the random series (X_i) is stationary.
- Symbolic representation of a sequence of consecutive prices: for example $(X_1, X_2, X_3) = (0, 1, 1)$ represents a decrease followed by two daily price increases.
- For a given length $L < n$, 2^L sequences are possible, ordered e.g. with Gray's binary code. If $L = 3$, the 8 sequences are $(G_1^3, ..., G_8^3) = ((0,0,0), (0,0,1), (0,1,1), (0,1,0), (1,1,0), (1,0,0), (1,0,1), (1,1,1))$.
- The probability to draw a particular sequence of length L is $p_i^L = \mathbb{P}((X_\cdot, X_{\cdot+1}, ..., X_{\cdot+L-1}) = G_i^L)$.
- For the probabilistic choice system $(\{0,1\}^{2^L}, (p_i^L)_i)$, the **Shannon entropy** is the amount of uncertainty in $(\{0,1\}^{2^L}, (p_i^L)_i)$ and is defined by:

$$H^L = -\sum_{i=1}^{2^L} p_i^L \log_2\left(p_i^L\right).$$

... and to market information

- We decompose a sequence $(X_1, ..., X_{L+1})$ (observed in time L) in:
  - an observed prefix $(X_1, ..., X_L) = G_i^L$ of probability p_i^L,
  - a random suffix X_{L+1} whose conditional distribution is Bernoulli of parameter $\pi_i^L \in [0, 1]$.

  The sequence $(X_1, ..., X_{L+1})$ is thus equal to $(G_i^L, 1)$ with probability $p_i^L \pi_i^L$ and $(G_i^L, 0)$ with probability $p_i^L(1 - \pi_i^L)$. The corresponding entropy is:

$$H^{L+1} = -\sum_{i=1}^{2^L} \left(p_i^L \pi_i^L \log_2 \left(p_i^L \pi_i^L \right) + p_i^L (1 - \pi_i^L) \log_2 \left(p_i^L (1 - \pi_i^L) \right) \right).$$

- The EMH (efficient market hypothesis) asserts that, conditionally on the prefix $(X_1, ..., X_L)$, the values of suffix $X_{L+1} = 1$ and $X_{L+1} = 0$ are equiprobable. As a particular case of the above with $\pi_i^L = 1/2$, the entropy according to the EMH is thus:

$$H_\star^{L+1} = -\sum_{i=1}^{2^L} p_i^L \log_2 \left(\frac{p_i^L}{2} \right) = 1 + H^L.$$

- We define the **market information** as the difference between the entropy according to the ideal EMH and the entropy of the market:

$$I^{L+1} = H_\star^{L+1} - H^{L+1}.$$

A statistical test of zero dependence

- The theoretical value of the market information I^{L+1} is 0 if and only if the market is efficient.
- As soon as EMH does not hold, we have $I^{L+1} > 0$.
- We evaluate empirically the probabilities $p_1^L, ..., p_{2^L}^L$ and $\pi_1^L, ..., \pi_{2^L}^L$ and get the empirical market information $\widehat{I_n^{L+1}}$.
- We still have $\widehat{I_n^{L+1}} \geq 0$, but if the EMH holds, we may have $\widehat{I_n^{L+1}}$ *slightly* higher than 0. Building a statistical test is thus mandatory for answering the question of the efficiency of the market.

Statistical test

(H_0) $I^{L+1} = 0$,

(H_1) $I^{L+1} \neq 0$.

Under (H_0), $\widehat{I_n^{L+1}}$ follows a gamma distribution $\Gamma(k, \theta)$ of shape parameter $k = 2^{L-1}$ and scale parameter $\theta = 1/\ln(2)\, n$.

The pdf of a gamma distribution is, for $x > 0$, $f(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}$, where $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, \mathrm{d}t$.

Distribution of serial information under (H_0)

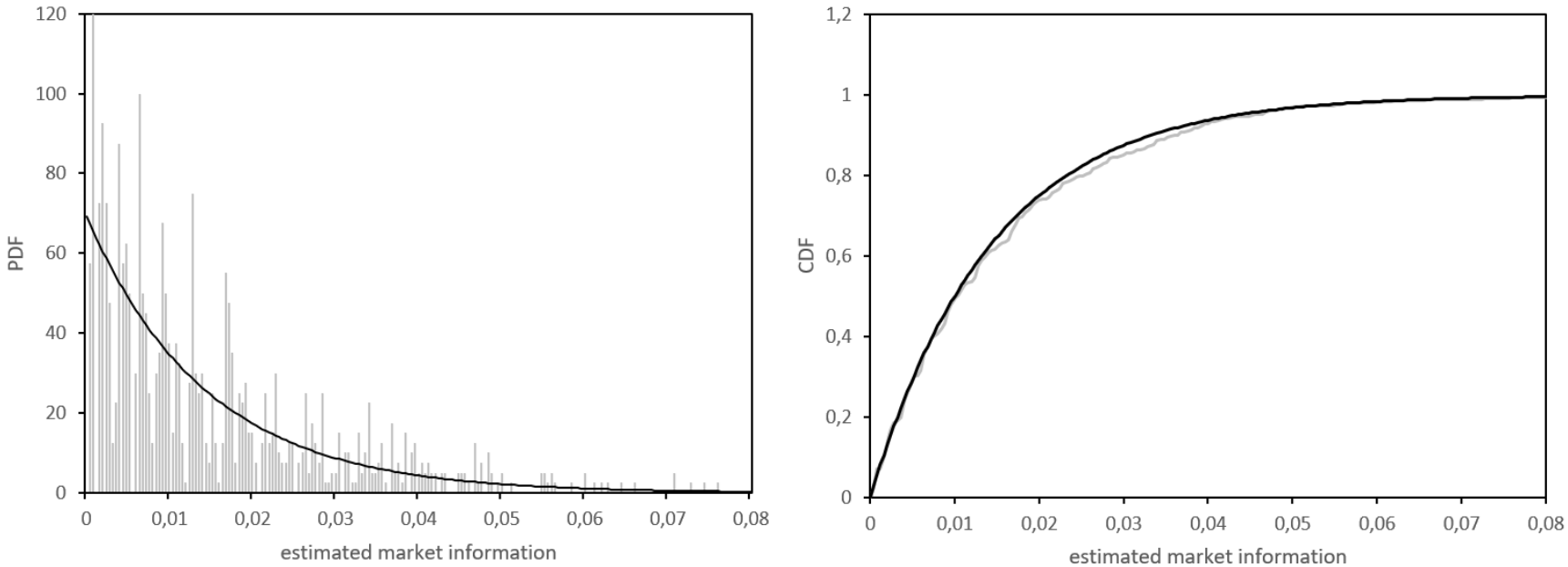

Figure: Asymptotic (black) and simulated (on 1,000 trajectories, grey) distributions of the estimated market information $\widehat{I_n^2}$ for $n = 100$.

Distribution of serial information under (H_0)

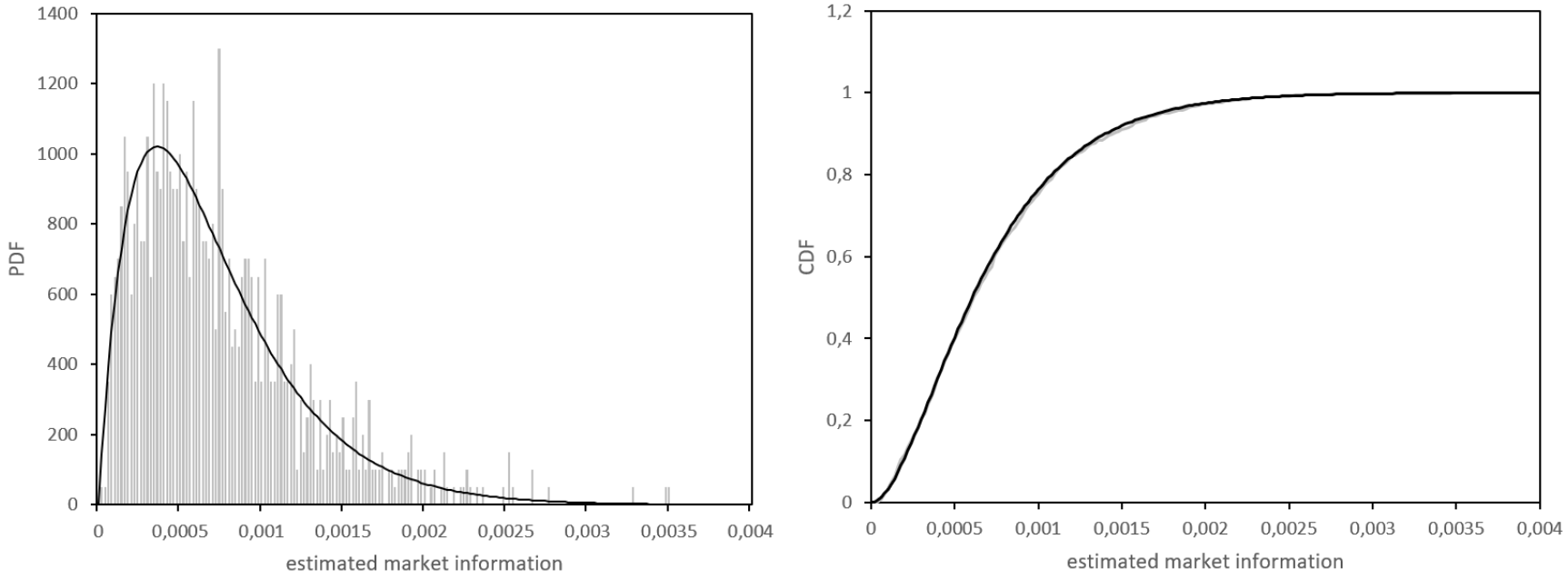

Figure: Asymptotic (black) and simulated (on 1,000 trajectories, grey) distributions of the estimated market information $\widehat{I}_n^3$ for $n = 4,000$.

Empirical analysis

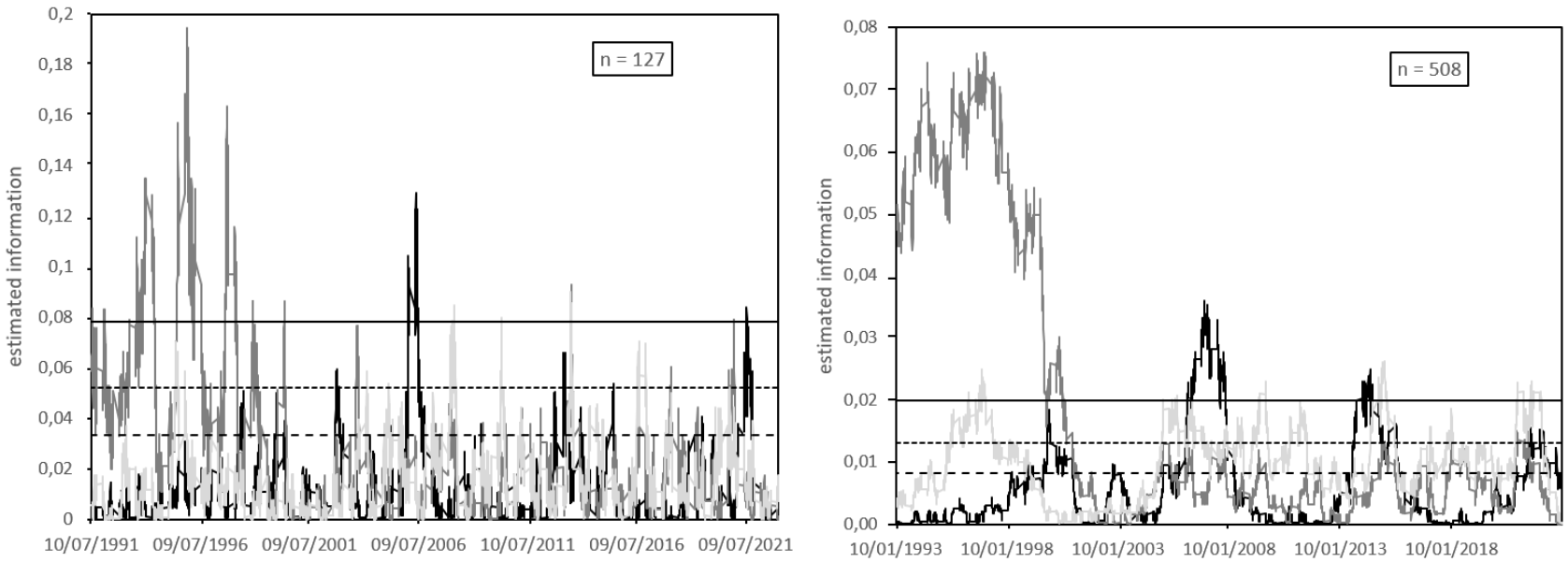

Figure: Estimated information $\widehat{I_n^2}$ in a rolling window of $n \in \{127, 508\}$ business days, for the CAC 40 index (black), Russell 2000 index (dark grey), Russell 3000 index (light grey), along with the confidence intervals of the statistical test of absence of information, with probabilities of 95%, 99%, and 99.9%.

Empirical analysis

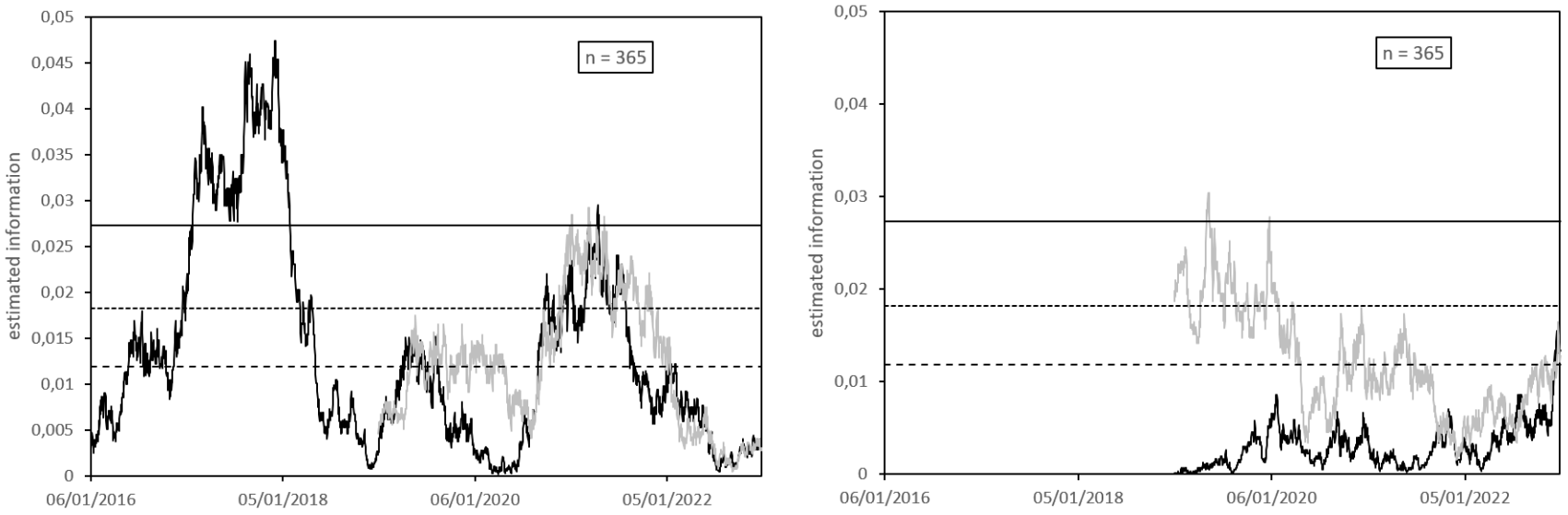

Figure: Estimated information $\widehat{I_n^2}$ in a rolling window of $n = 365$ days, for Bitcoin (left, black), Ethereum (left, grey), Ripple (right, grey), and Tezos (right, black), along with the confidence intervals of the statistical test of absence of information, with probabilities of 95%, 99%, and 99.9%.

$\rightsquigarrow$ Python exercise, Question 8.

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Some stylized facts on returns

Read for example *Cont, R. (2001), Empirical properties of asset returns: stylized facts and statistical issues*.

1. Heavy tails of price returns.
2. Gain/loss asymmetry: larger drawdowns than upward movements.
3. Volatility clustering.
4. Varying statistical properties through time scales.
5. Low autocorrelation of price returns.
6. Slow decay of autocorrelation of absolute price returns.
7. Leverage effect: volatility and price returns negatively correlated (for stocks at least).

$\rightsquigarrow$ Does your work in Python confirm these properties?

End of Part 1.